

SOLUTION QUIZ-03 BSE-6A

QUESTION-01

1. **Handshake:**

- A -> B: SYN=1, Seq=1200
- B -> A: SYN=1, ACK=1, Seq=4800, Ack=1201
- A -> B: SYN=0, ACK=1, Seq=1201, Ack=4801 (First data can piggyback here, usually considered the 3rd step).

2. **Data Segments:**

- Seg 1: Seq=1201 (Bytes 1201-2200).
- Seg 2: Seq=2201 (Bytes 2201-3200).

3. **Lost Segment 1:**

- If Seg 1 is delayed but Seg 2 arrives, TCP sends a **duplicate ACK** for the *expected* byte.
- **Ack = 1201.** (It acknowledges receipt of everything *up to* 1201, signaling it is still waiting for 1201).

QUESTION-02

(b) **Sequence number of lost segment: 700**

Table 1 – Complete Solution

TXNR (RTT)	Recv Window (rwnd MSS)	Cong.W window (cwnd MSS)	ssthresh (MSS)	RecvACK No.	Seq No(s) Transmitted	Timer (ms)	No Loss Or Timeout Or 3 dupACK	TCP Mode (SS/CA/FR)
0	8	1	8	N/A	N/A	600	No Loss	N/A
1	8	1	8	N/A	0	600	No Loss	SS
2	8	2	8	100	100, 200	600	No Loss	SS
3	8	4	8	300	300, 400, 500, 600	600	No Loss	SS
4	8	8	8	700	700, 800, 900, 1000, 1100, 1200, 1300, 1400	600	3dupACK	SS
5	8	7	4	700 (dup)	700 (retx)	600	3dupACK	FR
6	8	4	4	800	800, 900, 1000, 1100	600	No Loss	CA
7	8	5	4	1200	1200, 1300, 1400, 1500, 1600	600	No Loss	CA